GPT-3

Language Models are Few-Shot Learners

Brown et al. | arXiv:2005.14165v4

Scale one autoregressive language model to 175B parameters so many tasks can be specified through natural-language context rather than weight updates.

in-context learning · 175B parameters · few-shot

1 Problem

- Traditional NLP systems need large labeled datasets and task-specific fine-tuning, risking narrow specialization and leakage.
- Humans can infer a task from a description or a few examples; language models should approach the same interface.

2 Scale

125M TO 175B PARAMETERS

- A family of autoregressive Transformers shares one next-token objective; the largest has 96 layers and a 2048-token context.
- Performance follows smooth scaling trends, while qualitatively new few-shot behavior strengthens at the largest sizes.

3 In-context learning

- Zero-shot gives only instructions, one-shot adds one example, and few-shot packs demonstrations into the context.
- The model completes the pattern autoregressively; no gradient update or task-specific parameters are stored.

4 Pretraining

BROAD WEB + BOOK CORPUS

- Filtered Common Crawl, WebText2, two book corpora and Wikipedia form roughly 300B sampled training tokens.
- Higher-quality sources are sampled more often; contamination analysis estimates overlap with evaluation sets.

5 Evidence

42 LANGUAGE BENCHMARKS

- Few-shot GPT-3 is strong on translation, question answering, cloze tests and some reasoning tasks, often approaching fine-tuned systems.
- On LAMBADA it reaches 86.4% few-shot accuracy; larger models sharply improve arithmetic and novel-word tasks.

6 Meaning & limits

- GPT-3 reframed prompts as temporary task specifications and established in-context learning as a scaling phenomenon.
- It remains unreliable at reasoning, can memorize or emit toxic bias, and required about 3.14e23 training FLOPs.